

Ashish Kumar
 **Technical Lead – Backend & AI Integration Expert**

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Summary

- **Technical Lead & AI Engineer** with **16+ years** of experience in designing and delivering **scalable backend systems** and **AI-powered automation** solutions.
- Leading enterprise-scale projects at **Blink Charging**, including **EV CRM (Salesforce)**, **real-time WebSocket systems**, and **AI/Agentic AI integrations** using **LangChain** and **CrewAI**.
- **Proficient in backend technologies** — **Node.js, NestJS, Python**, and **microservices architecture** with **Kafka, Redis**, and **AWS (EC2, S3, Redshift, DynamoDB)**.
- Skilled in **ETL pipeline design, data orchestration**, and **serverless architectures** for high-performance and reliability.
- Experienced in **Generative AI, LangChain, CrewAI**, and **LLM orchestration** for building **AI-driven chatbots and autonomous agents**.
- Strong understanding of **system design, distributed systems**, and **real-time communication frameworks** (WebSockets, SSE).
- Adept in **containerization and DevOps** using **Docker, Kubernetes**, and **CI/CD pipelines**.
- Recognized for **driving innovation, mentoring engineering teams**, and delivering **enterprise-ready software ecosystems** with measurable business impact.

Key Skills

Languages & Frameworks

- JavaScript, TypeScript, Python
- Node.js, NestJS, ExpressJS, Django, LoopBack4, FastAPI
- React.js, Next.js

Architecture & Integration

- Microservices Architecture, Serverless, REST APIs, WebSockets, ETL Pipelines, System Design
- AWS (EC2, S3, DynamoDB, Redshift, DocumentDB), Kafka, Docker, Kubernetes, Redis

AI/ML & Generative AI

- LangChain, LangGraphCrewAI, Gradio, FastAPI, GenAI Chatbots, Agentic Workflows, Claude Code

Databases

- PostgreSQL, MongoDB, MySQL, MSSQL, DynamoDB, Amazon Redshift

DevOps & Tools

- Git, GitHub, GitLab, Bitbucket, Docker Compose, Jira, Power BI/Reveal BI
- NPM package development, Cron Jobs, JWT, Redis, Throttler

Core Competencies

Engineering Skills

- Backend Development with **Node.js & NestJS** (Modules, Providers, Guards, Interceptors, Middleware)
- RESTful API Design with **secure, scalable, and versioned contracts**
- **OAuth 2.0 / OpenID Connect (OIDC)** Authentication
- **Azure Active Directory (AAD)** Integration
- **Role-Based Access Control (RBAC)** & Multi-Application Authorization
- **PostgreSQL** Data Modelling, Query Optimization & Schema Design
- **Third-Party Service Decoupling** & Vendor Lock-in Reduction
- **Backward-Compatible System Migration** & Platform Modernization
- Clean Architecture, Modular Design & Dependency Abstraction
- Distributed Systems & Scalable Backend Architecture
- **Generative AI & LLM Integration (OpenAI APIs)**
- Prompt Engineering, Prompt Validation & Response Controls
- **AI Safety, Governance, Rate Limiting & Audit Logging**
- Secure AI Feature Enablement in Enterprise Applications
- Containerization & Cloud-Native Development (**Docker, Kubernetes**)

AI & Generative Tech

- LangChain, CrewAI, FastAPI for GenAI agents
- Prompt Engineering
- Conversational AI Design
- Gradio UI for agent front-ends

Soft Skills

- Strong problem-solving and debugging mindset
- Proactive decision-maker
- Clear communicator with distributed and local teams
- Leadership experience with cross-functional collaboration
- Experience leading client calls and sprint reviews

Work Experience

◆ **Node Js Developer (High Level and Low Level System Design)**– December 2025 to Present
VML Enterprise Solution -WPP Group (Previous Wunderman Thompson Commerce Private Limited), Noida, Uttar Pradesh

Project: Enterprise Multi-Application Access Management Platform

- Leading the **design and development of a greenfield backend platform using NestJS and TypeScript**, following clean architecture and enterprise-grade design patterns.
- Architecting a **new system version to decouple core business logic from a legacy third-party service**, while **preserving existing application behavior** to ensure smooth migration and zero functional regression.
- Implemented **Azure Active Directory (Azure AD) OAuth 2.0 authentication** with secure token validation and enterprise-compliant authorization flows.
- Designed and implemented a **Role-Based Access Control (RBAC)** framework mapping **Azure AD roles** to internal users, permissions, and multiple applications within a unified portal.
- Built a **multi-application access platform**, enabling users to dynamically access internal applications based on assigned roles and entitlements.
- Designed and optimized **PostgreSQL database schemas** for users, roles, applications, permissions, and audit data with a focus on scalability and data integrity.

- Developed secure, reusable, and well-documented **REST APIs** using NestJS best practices (modules, guards, interceptors, middleware).
- Integrated **OpenAI APIs** to enable **AI-generated content** for users, ensuring secure API usage, prompt control, and scalable request handling.
- Designed the AI integration in a **loosely coupled manner**, allowing future extensibility for additional AI models or providers.
- Ensured **enterprise security and compliance**, including token validation, role-based authorization, and protected AI endpoints.
- Collaborated closely with frontend teams, solution architects, and client stakeholders to align backend, security, and AI capabilities with business requirements.
- Implemented **enterprise-grade AI safety and governance controls** for OpenAI integrations, including prompt constraints, input/output validation, access control, rate limiting, and audit logging to ensure secure, compliant, and responsible AI usage.
- Contributed to **technical planning, architectural decisions, and migration strategy** for the next-generation platform.

Tech Stack: NestJS, Node.js, TypeScript, PostgreSQL, Azure AD OAuth 2.0, REST APIs, RBAC, Enterprise Security Architecture

◆ **Senior Software Engineer**– August 2024 to November 2025 **Blink Charging Software Pvt. Ltd.**, Noida, Uttar Pradesh

Projects: EV Station Platform, AI Chatbot System, ETL Pipelines, CRM & Reporting

- **Led architecture and development** of a full-scale EV charging station platform using microservices with modular services for CRM, real-time charger control, and analytics.
- Designed and built scalable **WebSocket infrastructure** using Socket.io & WS for live charger communication and remote control.
- Designed and implemented high-performance **ETL pipelines** across PostgreSQL, MongoDB, and DynamoDB with integration into Amazon Redshift and automated reporting on S3.
- Created dynamic **PDF/CSV reporting** solutions with advanced filtering capabilities.
- Led and executed **Generative & Agentic AI chatbot** systems using LangChain, FastAPI, and CrewAI, enhancing customer experience and automating support.
- Implemented a **custom URL shortening service** with Redis caching, click tracking, and REST APIs.
- Oversaw deployments using Docker/Docker Compose and initiated a **migration to Kubernetes** for better scalability.

Tech Stack: Node.js, NestJS, TypeScript, FastAPI, WebSockets, PostgreSQL, MongoDB, Redis, Amazon Redshift, AWS (S3, DynamoDB), Kafka, LangChain, CrewAI, Docker, Kubernetes

◆ **Technical Lead | Senior Software Developer**– November 2021 to July 2024 **GIIT Solutions**, Gurugram, Haryana

Project: EV Charging Station Platform

- Engineered and managed a complete back-end system with **microservices architecture** for user/host portals, billing, reporting, and ETL.
- Integrated **Reveal BI dashboards**, replacing Power BI with heavy customizations for advanced visualization.
- Designed and implemented and automated **cron jobs** for monthly host/user statements, invoice uploads to S3, and Redshift-based analytics.
- Led DevOps effort with **Dockerized environments** and Kubernetes readiness.

Project: Healthcare Portal (US Client)

- Migrated legacy PHP services to **Node.js (LoopBack4)** with secure API integrations for doctors and patients.
- Integrated device APIs and built questionnaire-based recommendation logic.
- Created **EJS/Handlebars-based dynamic PDF** generators.

Project: College Admission Portal

- Delivered API suite for registration, authentication, and appointment scheduling.

- Upgraded Express.js legacy services to **NestJS** with MSSQL.
- Collaborated with front-end teams on **Next.js-based** admin and parent portals.

Project: DevOps Plugin Platform (Israel Client)

- Built **cloud-native SDK-based plugins** for AWS, GCP, Azure to integrate with CI/CD pipelines.
- Designed and implemented back-end plugins in Node.js with Express; coordinated Vue.js-based front-end integration.

Tech Stack (across projects): Node.js, NestJS, LoopBack4, Express, PostgreSQL, MongoDB, DynamoDB, MySQL, MSSQL, Redshift, AWS, Docker, Kubernetes, Vue.js, React.js, EJS, Handlebars, JWT, Jira

◆ **Software Developer**— June 2019 to November 2021 **Tag11 Infotech Pvt. Ltd, Jaipur, Rajasthan**

Project: Wealth Bazaar

- Built a mutual fund transaction platform with SEBI-verified APIs.
- Designed and implemented complete user journey from onboarding to transactions with reporting and history logs.

Project: Banking Software

- Delivered core banking features (accounts, transactions) with Express.js and PostgreSQL.
- Created a clean, responsive UI using Bootstrap, HTML, and Vanilla JS.

Tech Stack: Node.js, Express, NestJS, PostgreSQL, MySQL, MSSQL, HTML/CSS, Bootstrap, JavaScript

◆ **Freelancer**— December 2010 to May 2019 **New Delhi, India**

Project: Online Beer Store (Canada)

- Designed and implemented a scalable e-commerce backend using Express.js and NestJS.
- Implemented **store locator, session-based JWT authentication**, and real-time inventory syncing.

Project: School ERP (Colegios)

- The operational platform a school actually runs on — admissions, academics, staff, parents and finance — with AI and agent tooling designed into the workflow instead of parked in a side panel..
- Architected a modern school ERP covering the full operational surface of an institution, built AI-first rather than retrofitted.
- Designed the multi-tenant data model so each school is isolated while the platform stays a single deployable product.

Tech Stack: Node.js, Express, NestJS, PostgreSQL, MySQL, MSSQL, HTML/CSS, Bootstrap, JavaScript

Education

B.E. – Computer Science (2006–2010)

Institution of Electronic and Telecommunication Engineers, New Delhi

CBSE – 12th & 10th Standard

Delhi

Languages

English, Hindi

Certifications

- Advanced JavaScript – Cutshort (ID: 51737)
- Advanced NodeJs – Cutshort (ID: 60850)
- Python – Cutshort (ID: 49866)
- Generative AI Foundation Certificate Program- Upgrade in collaboration with Microsoft (UID: LVM6yuUQAKSa8AWY) [Upgrade Certificate](#)
- HTML/CSS – Cutshort (ID: 50040)

Awards & Achievements

- Lifetime Associate Member – IETE (Institution of Electronics and Telecommunication Engineers)
- 3-Star Rating – HackerRank
- Contributor – Dev.to, LinkedIn Articles, Medium.com

Hobbies

Exploring tech, coding, reading PMA (Positive Mental Attitude) books, Technical communities contribution

Profile Links

- [LinkedIn](#)
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- [Dev.to](#)
- [Medium.com](#)
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